

Muhammad Hafiz

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Computer Engineering graduate with 2+ years of research experience in machine learning, deep learning, computer vision, data preparation, and model evaluation. Built and shipped a football recruitment intelligence system that transforms event data into player profiles, similarity search, and explainable recruitment rankings. Co-author of an IEEE CENIM paper and experienced with Python-based ML, NLP/retrieval, data pipelines, and reproducible evaluation.

EDUCATION

Universitas Syiah Kuala	Banda Aceh, Indonesia
Bachelor of Computer Engineering (S.T.), GPA: 3.49/4.00	2019–2024
Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures and Algorithms, Software Engineering.	

RESEARCH EXPERIENCE

Research Assistant	May 2022–Nov 2024
<i>Universitas Syiah Kuala, Banda Aceh, Indonesia</i>	
- Developed and evaluated three-class breast thermal-image classification using VGG16/ResNet50 feature extraction and XGBoost, applying deep learning, computer vision, and pattern-recognition workflows. - Curated a balanced 1,842-image dataset and built preprocessing, model-comparison, hyperparameter-tuning, and evaluation pipelines in Python. - Documented experimental results and limitations, collaborated with academic researchers, and co-authored an IEEE CENIM 2024 paper. IEEE Xplore.	

SELECTED PROJECTS

Football Recruitment Intelligence Engine V1	Python / Football Analytics
- Built an end-to-end pipeline using StatsBomb Open Data to produce quality-gated player profiles, positional peer normalization, player similarity retrieval, and explainable weighted recruitment rankings. - Implemented canonical data contracts, data-quality validation, feature engineering, reproducible builds, a local FastAPI application, and 226 automated tests; preserved uncertain observations with explicit quality flags instead of fabricating corrections. - Published source code and technical case study as a portfolio project. GitHub repository.	
RAG CVE System with Privacy Protection	Python / NLP
Built a CVE retrieval pipeline with sentence-transformers, ChromaDB, and Ollama; implemented configurable Groq/OpenAI client code and output-level PII checks. Code repository.	

ADDITIONAL EXPERIENCE

Machine Learning Student, Bangkit Academy	Feb–Jul 2022
Completed the Machine Learning track and contributed to BADI (Business Analytics Daily Income), a Python-based income-prediction and business-analysis project.	
Internship Trainee, Telkom Indonesia	Jul–Aug 2022
Worked with the Integrated Operation Center on network operations, including IPv6, remote ticketing, and assurance systems.	

TECHNICAL SKILLS

AI / Machine Learning: Machine Learning, Deep Learning, Neural Networks, Computer Vision, NLP, Pattern Recognition, Data Mining, model evaluation, hyperparameter tuning.

Programming / Data Engineering: Python, SQL, Pandas, NumPy, scikit-learn, XGBoost, data pipelines, feature engineering, schema/data validation.

Frameworks / Systems: TensorFlow/Keras, VGG16, ResNet50, FastAPI, Pydantic, sentence-transformers, ChromaDB, Ollama.

Tools: Git, GitHub, pytest, Jupyter Notebook, Google Colab, uv, Ruff.

TRAINING

TensorFlow Developer Certificate (2022); Bangkit Academy Machine Learning track (2022).